



Assisting consumers in detecting fake reviews: The role of identity information disclosure and consensus



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ABSTRACT

Current discussions in academia and in the press increase consumers' awareness of potentially deceptive online reviews. The increasing practice of fake reviews posted online not only jeopardizes the credibility of review sites as important information sources for individuals but also endangers a valuable source of information for service providers. Two studies shed further light on the role of consensus and identity-related information in assisting consumers detect potentially faked reviews. In one preliminary study, a sample of 4826 rejected and 4881 published online reviews was analyzed to investigate the differences in the disclosure of author-related information such as name and age as well as star ratings across those reviews. In the main study, a 3 (identity disclosure) $\times$ 2 (consensus) $\times$ 2 (priming of fake reviews) experiment was carried out with 390 respondents. The results highlight the relevance of the review's consensus in relation to the overall rating of previous reviews and corroborate the results of the preliminary study from the perspective of an internet user: the value of the amount of available information on the review's author in assisting individuals detect potential fake reviews. This study complements research in computer science by highlighting the relevance of contextual—in addition to textual—indicators that assist internet users in detecting potentially deceptive online reviews.

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1. Introduction

In a recent post, Forbes contributor Cheryl Conner (2013) discusses the bright and dark sides of a sector that has become a \$5 billion industry: online reputation management (ORM). As consumers increasingly interact with each other over the internet and exchange opinions and experiences on various service encounters, firms are increasingly devoting resources to monitoring, analyzing, repairing, and improving their reputations online. While many ORM firms are legitimate, providers are increasingly adopting deceptive practices, such as publishing bogus reviews. The literature refers to the practice of fake reviews—or deceptive opinion spam—as “fictitious reviews that have been deliberately written to sound authentic, to deceive the reader” (Ott, Cardie and Hancock, 2012, p. 201); estimations hold that up to one-third of all online reviews are fake (Streitfeld, 2012). Those alarming numbers and the considerable threat to the trustworthiness of review sites for consumers have led to several initiatives by regulatory institutions, review sites, and independent associations to fight the increasing prevalence of deceptive opinion spam. In September 2013, for example, 19 companies that engaged in generating and

publishing fake reviews on a New York-based fictitious yoghurt shop were fined a total of \$350,000 (Rushe, 2013). In October 2015, Amazon announced it would sue more than 1000 fake reviewers in a Seattle lawsuit (Gani, 2015). From a broader perspective, the World Economic Forum (WEC) identifies digital wildfires in a hyperconnected world as a timely major concern in its 2013 Global Risks report (Howell, 2013): characteristics such as anonymity, speed, and scale of online communication could “enable the rapid viral spread of information that is either intentionally or unintentionally misleading [...], with serious consequences” (Howell, 2013, p. 23) and deception (i.e., intentional) or misinformation (i.e., unintentional) can be deliberately propagated by those “who stand and reap some kind of benefits” (Howell, 2013, p. 25).

The perspective on different initiatives to report and fight against deceptive practices on the internet in the form of fake reviews provides evidence for the pervasiveness of opinion spam. Scholars from computer science recently increased their efforts to distinguish features of fake reviews as opposed to truthful ones. The identified characteristics included temporal patterns (Xie et al., 2012), content-related aspects such as word use (Ott et al., 2011), word duplication and number of sentences (Bhattacharai, Rus and Dasgupta, 2009), and different reviewer- and behavior-related characteristics based on machine-learning methods (Jindal and Liu, 2008; Li et al., 2011; Lim et al., 2010). Ott and his colleagues

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developed algorithms and models based on their findings, transferring knowledge from linguistics to fake review detection, and published the tool Review Skeptic (www.reviewskeptic.com). Internet users can copy and paste hotel reviews on the website to test whether the review is real or fake—with 90% accuracy.

Where these sophisticated approaches from computer science are extremely helpful for review site operators to update and improve their opinion spam detection algorithms, ORM firms are able to adapt to these recent findings and include them with the aim of generating fake—but authentic sounding—reviews for their clients.

While numerous institutions and associations attempt to reduce the number of fake reviews and fine deceptive behaviors, a considerable amount of deceptive opinion spam is still present today—Yelp recently admitted that up to 25% of reviews on its site are at least suspicious (Roberts, 2013)—which highlights the importance for consumers to develop skills to detect potentially deceitful reviews. However, whereas consumers adopt a deception-aware mindset in other market-related contexts and are aware of persuasion attempts by marketers in the case of traditional advertising, they do not adopt this mindset in the context of online reviews. As internet users expect to interact with other consumers in the customer-to-customer environment of review sites, they usually do not wear the same shields of self-protection and skepticism.

Consequently, the objective of the present study is to complement recent findings from related disciplines that adopt review-centric approaches from a consumer-centric perspective. To do so, different cues that potentially assist readers of online reviews in detecting potentially deceptive opinion spam are investigated. The amount of available information on the author of a review is a potential additional cue that helps internet users shape their perception of the review author's credibility. The disclosure of information on the author of a review has attracted interest from scholars investigating the effects of covert persuasion attempts (Campbell, Mohr and Verlegh, 2013). A large sample of 9707 reviews submitted to a multi-services review site was analyzed in an initial study. The sample contained two types of reviews: a first set of reviews ($n = 4826$) that were rejected by the site manager based on text-related reasons such as the length of the text (i.e., too short) or the inclusion of inappropriate terms. Human detectors performed the assessment of those reasons. A second set of reviews ($n = 4881$) underwent verification by site managers and were published on the review site. The results show that binary factors related to the author's identity (i.e., disclosure of age, name, and phone number) as well as the amount of disclosed identity-related information significantly separated the two sub-samples such that reviews that were actually published included a higher level of identity-disclosure than rejected reviews.

Based on the findings of the preliminary study that underline the practical relevance and validity of the cue, a second study was conducted to experimentally investigate the role of identity-related information about the review's author in readers' assessments of its trustworthiness. In addition to the available information about the review's author, two additional potential cues were investigated. Inspired by the theoretical underpinnings of the Persuasion Knowledge Model (Friestad and Wright, 1994) as well as increasing media coverage informing consumers about deceptive practices in the form of fake reviews (Tuttle, 2012), an activation of the deception-aware mindset through exposure to a news article on fake reviews was included. Second, the consensus of a reviewer's opinion with the majority is increasingly given as an indicator by the press and consumer associations to differentiate true from deceptive reviews (Johnston, 2013). Additionally, consensus information in online reviews is also attracting increasing attention from marketing research (Benedickus, 2011).

The findings highlight the importance of consensus information on the perceived trustworthiness of a review's source. Furthermore, the disclosure of information about the reviewer influences the perceived trustworthiness directly and is also moderated by consensus. Finally, the investigation of potential mediation of different deception detection cues on behavioral intentions—the individual's intention to purchase as well as his or her intention to avoid the service provider in the future—through the source's trustworthiness further supports the relevance of trust in public online environments such as review sites.

The present study contributes to the marketing literature in several ways. First, the effects of available cues to evaluate the trustworthiness of online review contributors are empirically assessed. Second, the findings complement research in computer science by adopting a consumer-centric perspective in the evaluation of online reviews. Thus far, the marketing discipline has largely neglected research on deceptive practices in the form of fake reviews. Third, the results show the relevance of perceived trustworthiness as a mediator and its effects on positively connoted outcomes (i.e., purchase intention) and negative variables (i.e., avoidance behavior). The present paper further calls for future interdisciplinary research on the effects of deceptive communication in online environments to motivate service providers to cease and desist from such malicious practices.

2. Background and hypothesis development

2.1. Persuasion and deceptive communication in online environments

The internet has not only added new potential ways for marketers to communicate to and interact with their customers, it also enables consumers to exchange information, experiences, and opinions in various domains with fellow consumers. The consequence of this opportunity to conveniently acquire credible information from sources that are at least perceived to have no commercial interest in recommending brands, services, products, and companies, is bad news for marketers: individuals trust peers much more than marketing-related sources such as traditional or digital advertising efforts (Nielsen, 2013). They even tend to avoid or block out firms' persuasion attempts via marketing communications (Kaikati and Kaikati 2004; Kimmel, 2010; Rumbo 2002). Research on today's pervasive forms of customer interactions online—usually referred to as electronic word-of-mouth (eWOM)—confirms its effects on customer behavior and attitudes (Gupta and Harris, 2010; Senecal and Nantel 2004; Zhu and Zhang 2010). However, the power of positive online recommendations from peer consumers, the potential threat of negative articulations posted in public online environments, and interactions that are largely characterized by anonymity between interlocutors have led to numerous deception attempts by marketers (Magnini, 2011; Xiao and Benbasat 2011). In an aim to circumvent consumers' avoidance and tendency to block out marketing communications, marketers pose as prior customers and publish fake positive reviews on their own services (Darke and Ritchie, 2007; Xiao and Benbasat 2011). This form of online deception fulfills the three main conditions of deceptive communication as it is currently defined (Masip, Garrido and Herrero, 2004). First, the generation and publication of fake reviews can be considered a deliberate act by the marketer and can therefore be distinguished from misinformation that is usually understood as non-intentional distortion of the truth (Xiao and Benbasat, 2011). Second, fake online reviews are the result of manipulation of information about the marketplace by impersonating a usual customer (Boush, Friestad and Wright, 2009; Xiao and Benbasat, 2011). Third, fake positive

reviews are posted by marketers to influence and attract potential customers and, hence, are goal-oriented (Xiao and Benbasat, 2011). Without this deceitful information, individuals—or the victims of deceptive opinion spam—would unlikely take the same action or hold the same belief about the service or service provider (Grazioli and Jarvenpaa, 2003). The major concern with this rising practice of covert, masked, or stealth marketing (Campbell, Mohr and Verleghe, 2013; Kaikati and Kaikati, 2004) lies within the fact that individuals do not activate the same persuasion knowledge and skepticism, as they expect to read experiences and opinions from peer consumers and not marketers (Campbell, Mohr and Verleghe, 2013; Darke and Ritchie 2007).

In the following, a number of potential cues are discussed to make deceptive intent in online reviews salient (Campbell and Kirmani, 2000; Kirmani and Zhu 2007). As recent empirical findings in online environments provide support for the importance of trust-related factors in consumer decisions and behavioral intentions in the internet (Bart et al., 2005; Benedicktus et al., 2010; Schlosser, White and Lloyd, 2006), the trustworthiness of the reviewer's source is an essential contextual factor in the present study. While trust refers to "an expectancy held by an individual (...) that the word, promise, verbal or written statement of another individual (...) can be relied upon" (Rotter, 1967, p. 651), trustworthiness presents an essential basis or antecedent of trust (Colquitt, Scott and LePine, 2007; Mayer, Davis and Schoorman 1995). A source's trustworthiness can be defined as "the degree of confidence in the communicator's intent to communicate the assertions he considers most valid" (Hovland, Janis and Kelley, 1953, p. 21).

2.2. Disclosure of identity-descriptive information

As opposed to traditional word-of-mouth (WOM) communication that takes place face-to-face (Arndt, 1967; Sun et al., 2006), its digital counterpart—electronic word-of-mouth (eWOM) (e.g., online reviews)—is characterized by asynchronized communication between complete internet users with limited information about each other (Schindler and Bickart, 2005). For review sites in which authors and readers of online reviews are complete strangers, the disclosure of identity-descriptive information can support readers in accurately deriving information about the reviewer's identity and motives (Forman, Ghose and Wiesenfeld, 2008; Mayzlin, Dover and Chevalier 2014). According to the information processing literature, attributes of the message source are directly associated with the recipient's attitudes and behaviors (Forman, Ghose and Wiesenfeld, 2008). Chaiken (1980) argues that recipients of information adopt a heuristic approach in which they rely more on immediately accessible information that is not content-related, such as the source's identity, as they tend to exert little effort in evaluating message validity. The disclosure of identity-descriptive information is usually a choice of the reviewer's author when posting a review and typically includes his or her real name, age, location, and consumption context.

Existing empirical research widely focuses on the relationship between the disclosure of identity information and the helpfulness ratings of online reviews. Ghose and Ipeirotis (2011) find significant effects of a reviewer's disclosure of identity-descriptive information (i.e., real name or location) on the helpfulness ratings of the review for other internet users and subsequent sales. Forman, Ghose and Wiesenfeld (2008) analyze reviews on Amazon.com and reveal a potential impact that the reviewer's disclosure of identity-descriptive information may have on sales. However, the disclosure of identity-descriptive information is treated as binary variable in both studies (Forman, Ghose and Wiesenfeld, 2008; Ghose and Ipeirotis 2011), which means the disclosure of the real name or location was analyzed as any disclosure. Similarly, Baek

et al. (2012) find a significant relationship between the disclosure of the reviewer's real name and the helpfulness of the review. Furthermore, Kusumasondjaja et al. (2012) conducted an experiment to show a significant main effect of the reviewer's disclosed information and the perceived credibility of the review: the disclosure of identity-descriptive information led to higher credibility than undisclosed information.

In the case of online reviews, the rationale is as follows: the more information about the author of the review is accessible for the recipient, the more he or she evaluates the review's source as trustworthy. Thus, we formulate the following hypothesis:

H1. The more identity-descriptive information is disclosed about the reviewer, the more trustworthy the review's source will be evaluated.

2.3. Consensus information

The second potential cue readers of online reviews use to derive the trustworthiness of its source—consensus information—has already received some attention from marketing research. Consensus information in the context of online reviews refers to the extent of perceived agreement among reviewers in their evaluation of a product or service (Jiménez and Mendoza, 2013) and can be expressed through the similarity of an evaluation with the average rating on the review site (Benedicktus et al., 2010). In the interpersonal persuasion literature, Cialdini (2007) refers to consensus information as social proof and underlines its importance in persuasion and influence attempts. In their recent work, Jiménez and Mendoza (2013) experimentally investigate the effects of detail of information in the review and reviewer agreement (i.e., consensus information) on purchase intention and show that the credibility of the source fully mediates this effect of consensus on purchase intention in the case of an experience good (i.e., hotel review; study 2). Furthermore, Benedicktus (2011) examined the effects of consensus. Finally, Benedicktus et al. (2010) researched the effects of consensus information via satisfaction ratings with retailers and provides support for the effects of consensus on the perceived trustworthiness of the retailer and the respondent's purchase intention. Additionally, Benedicktus and colleagues found that trustworthiness mediated the effect of consensus information on purchase intention.

Furthermore, from attribution (Kelley and Michela, 1980) and attributional theory (Weiner, 2000) we know that consensus information provides an orientation for the reader of a positive review: the reason for the positive reviews can either be attributed to the author of the review—and therefore can potentially be deceptive—or to the performance of the service provider. As information on the reviewer's identity and incentives are scarce (Mayzlin, Dover and Chevalier, 2014), consensus information potentially acts as additional information to correctly attribute the reasons for the positive review to the service experienced or to the review's author (Kelley and Michela 1980; Orvis, Cunningham and Kelley, 1975). This means that if the positive review is in line with the overall opinion of other reviewers on the review site, the positivity of the reviewer's experience will be attributed to the performance of the service provider. But, in the case of low consensus, the reviewer's opinion strongly differs from the general opinion and the reader might attribute the positivity to the source of the review instead of the service provider. Consensus information can therefore provide additional information to correctly attribute the causes for the positive review (Kelley and Michela 1980; Orvis, Cunningham and Kelley, 1975). Empirical research provides further support for the impact of consensus information on trustworthiness (Ba and Pavlou, 2002; Bolton, Katok and Ockenfels 2004; Pavlou and Dimoka 2006). Consequently, we expect that the

average rating provided by the usually considerable number of previous reviews is immediately accessible and potentially assists the internet user in making causal inferences about the review's source.

H2. The source of the online review is perceived as being more trustworthy when the consensus of the reviewer's opinion with the average rating of the service provider is high than when the consensus is low.

In addition to the main effect, it is expected that the effect of the amount of available information on the perceived trustworthiness will be moderated by the consensus information. The rationale for the expected interaction between the two information cues is as follows: the more reviewers disclose information about themselves, the more trustworthy they should be perceived by the reader of the review. However, the less information is available on the source itself, the more the internet user turns to the second cue, i.e., the extent to which the review in question is in line with the assessments by other reviewers. As a consequence, we expect that the review's source is still perceived as rather trustworthy in the high consensus case even though little information about the review's author is available. In turn, when the consistency with the grand average of reviews is low, trustworthiness declines substantially with the decrease of other remaining available cues: the disclosure of identity information about the review's source.

H3. The consensus information moderates the positive impact of the amount of disclosed information about the author of the review such that high consensus with the average assessment of the service provider compensates more for a lack in identity disclosure than the low consensus of the review with the overall assessment of other reviewers.

2.4. Activation of a deception mindset

Through their persuasion knowledge model, Friestad and Wright (1994; 1995; 1999) draw a picture of individuals as consumers of marketplace persuasion and highlight consumers' knowledge of persuasion attempts—including deceptive ones—as important in terms of actively and successfully coping with those strategies. Friestad and Wright (1994) establish a three-component belief system in which consumers process marketplace messages: marketplace persuasion knowledge (PK) that includes marketers' goals and tactics as well as the individual's own persuasion-coping knowledge and skills; agent knowledge (AK) that comprises the traits, competencies, and objectives a specific marketer who presents a message; and message topic knowledge (TK) that integrates beliefs about the specific object of the message. The authors later merge PK with an additional belief system—marketplace deception beliefs and knowledge (MDK)—which explicitly includes marketplace deception as persuasion attempts applied by agents (Boush, Friestad and Wright, 2009). Furthermore, consumers acquire their persuasion (and deception) knowledge from various interactions and sources such as their intimate peers, observation of marketing activities, or from news reports on marketing tactics (Bambauer-Sachse and Mangold, 2013; Friestad and Wright 1994). Empirical research shows that persuasion attempts in terms of price reductions (Hardesty, Bearden and Carlson, 2007), radio advertisement for a brand (Wei, Fischer and Main, 2008), and in-game-advertising (Lorenzon and Russell, 2012) lead to more negative evaluations of the offer or brand by individuals with a higher level of persuasion knowledge. Recently, Bambauer-Sachse and Mangold (2013) turned to the role of persuasion knowledge in the evaluation of online product reviews. The authors find some mixed results regarding the effects

of the activation of persuasion knowledge depending on the source of this knowledge and the mix of positive and negative reviews.

In the context of online reviews, an environment where strangers interact with each other and readers either expect a review's author to be a peer consumer might be uncertain if the review was written by an unbiased peer customer or a biased commercial source (Mayzlin, Dover and Chevalier, 2014). Consequently, the activation of persuasion knowledge directly concerns the source of the review or—in terms of the PKM—the beliefs of the agent's identity and objectives. Through the activation of persuasion knowledge in online review sites, individuals apply a deception-aware mindset (Boush, Friestad and Wright, 2009) that should entail suspicion about the motives of the review's author and a lower perception of the source's trustworthiness (Kirmani and Zhu, 2007). Thus, we expect the perceived trustworthiness of the sender to be lower when persuasion knowledge about online reviews is activated.

H4. When persuasion knowledge (i.e., the knowledge that online reviews can be manipulated and review sites lack of competence to filter fake reviews) is activated, individuals evaluate the review source as being less trustworthy than when persuasion knowledge is not activated.

2.5. Trustworthiness as a mediator between information cues and behavioral intentions

The literature largely supports the effect of eWOM via online reviews on behavioral intentions such as purchasing intentions (Chatterjee, 2001; Tsang and Prendergast 2009). However, few studies have investigated the effects of eWOM on credibility and trust as perceived by the receiver (Chatterjee, 2001; Senecal and Nantel 2004; Tsang and Prendergast 2009). This wide neglect of trust-related factors is surprising given that trust was identified as a crucial driver of online interactions and transactions (Lee and Turban, 2001; Peppers and Rogers 2012) and should become even more salient due to the recent discussion of deceptive reviews (Malbon, 2013; Mayzlin, Dover and Chevalier 2014). In their commitment-trust theory, Morgan and Hunt (1994) highlight the importance of trust as a mediating variable. Empirical research further supports the impact of the source's trustworthiness on behavioral intentions (Eisend, 2004; Wangenheim and Bayón 2007). In our research, we focus on two main dependent variables that we expect to be affected by the source's trustworthiness: the internet user's intention to purchase and intention to avoid the service provider in the future. While the purchase intention considers the user's approach behavior, we are particularly interested in avoidance behavior as an intentional withdrawal from doing business with the service provider in question (Graham Austin, Plouffe and Peters, 2005) and, thus, complement the individual's intention to patronize the reviewed service with some form of resistance. Consequently, we posit that the perceived trustworthiness of the source mediates the effects of the available cues (i.e., identity disclosure, consensus information) and the activation of persuasion knowledge on these two behavioral intentions.

H5. The effect of the sender's extent of disclosed identity information on (a) purchase intention and (b) avoidance behavior is mediated by the perceived trustworthiness of the review's source.

H6. The effect of consensus information on (a) purchase intention and (b) avoidance behavior is mediated by the perceived trustworthiness of the review's source.

H7. The effect of the consumer's activated persuasion knowledge on (a) purchase intention and (b) avoidance behavior is mediated

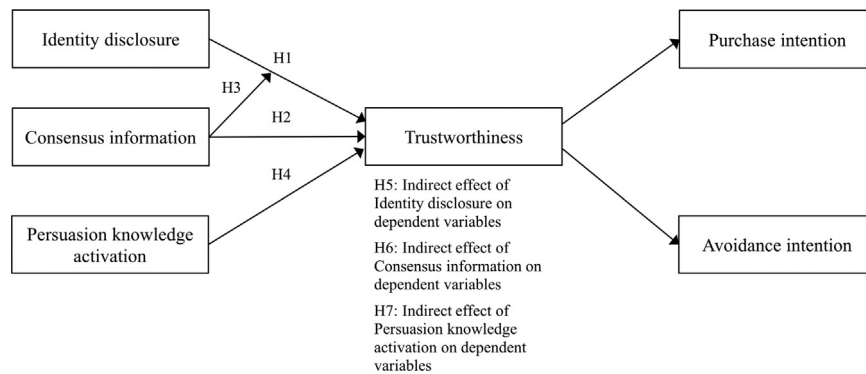


Fig. 1. Conceptual model.

by the perceived trustworthiness of the review's source.

Fig. 1 depicts the complete conceptual model and the formulated hypotheses.

3. Preliminary study

3.1. Purpose and data description

To better understand the raw material of the study – online reviews – and the relevance of the proposed research for different stakeholders such as review site managers, we collaborated with a large Western European review site where internet users can post their opinions and experiences with a large variety of services and products. Submitted reviews undergo a process of quality assessment by site collaborators who decide whether to publish the submitted review. The evaluation procedure focuses on the review text, and different text-related characteristics that increase the suspiciousness of the review were identified: those characteristics include the length of the review (i.e., if the review is too short, the review becomes more suspicious) and word use (e.g., inappropriate terms such as cursing or the use of commercial vocabulary).

The review site management provided a large dataset of two types of reviews. The first was a set of 4826 submitted reviews that were rejected after the quality check explained above. The second set consisted in a similar number of reviews (i.e., 4881) that passed the quality check and were published on the site. If provided by the internet user, the dataset included identity-descriptive information about the reviewer such as age (via the author's birth date), name, phone number, and email address. Furthermore, the data contained the title and text of the reviews as well as the assigned star rating between 0 and 5 where five stars was the most positive evaluation of the service in question by the reviewer.

3.2. Coding procedure

The different identity-descriptive elements were coded as binary variables (i.e., 1 'information provided', 0 'information not provided'), which is consistent to prior studies (Forman, Ghose and Wiesenfeld, 2008; Ghose and Ipeiritos 2011). The provision of age (i.e., birth date), name, and phone number are optional on the review site. Phone numbers were checked for consistency with standard formats (10 digits) and realistic (e.g., excluding number such as '111111111'). To assess the total amount of information provided by the internet user along with the review, we combined the three main identity-related descriptions in an indicator we labeled "identity disclosure level". The indicator was a simple

addition of the three types of identity information and, thus, varied between '0' (i.e., neither age, name, nor phone number were provided) and '3' (i.e., the internet user provided information about age, name, and phone number).

As the provision of a valid email address is compulsory for internet users who submit a review, a different coding procedure was adopted: if the domain of the provided email address was issued by a telecommunication provider, the value '1' was attributed. If the domain pointed at operators that issue free email accounts (e.g., hotmail.com, google.com), we attributed the value '0'. The rationale behind this coding was as follows: telecommunication providers assign email addresses only to their customers. Thus, the domain potentially reveals additional information about the real or bogus identity of the author of the submitted review. We further looked at text length and used the number of characters for our analyses. To do so, the reviews underwent a cleaning process to delete unnecessary spaces between words.

3.3. Results

A series of *t* tests in SPSS was conducted to assess potential mean differences between the two sub-samples (i.e., rejected and accepted online reviews). The results are presented in Table 1.

The results reveal several significant differences between the two sub-samples. First, the authors of reviews that were rejected after the quality check by the site manager based on the review text provided significantly ($t = 3.75$, $p < .001$) less information about their identity ($M = .89$, $SD = 1.22$) than did the authors of accepted reviews ($M = .98$, $SD = 1.24$). While the difference in terms of disclosed names is only significant at the 5% level ($t = 2.10$, $p = .035$), the differences for age ($t = 3.44$, $p = .001$) and phone number ($t = 4.92$, $p < .001$) are highly significant. In addition to those identity-related characteristics, the email provider type potentially serves review site operators as an additional indicator ($t = 11.50$, $p < .001$): while 45% of authors of accepted reviews provided an email address that relates to a telecommunication provider's domain, around only one third (i.e., 34%) of authors of rejected reviews provided such an email address.

The results of the additional information further show that texts of accepted reviews are significantly ($t = 19.96$, $p < .001$) longer ($M = 256.59$, $SD = 286.79$) than those of rejected reviews ($M = 149.20$, $SD = 241.61$). However, as the acceptance decisions of the site managers' were made based on the review text, this result might be less convincing and be interpreted with caution. Finally, the authors of rejected reviews evaluated the services and products significantly ($t = 8.80$, $p < .001$) better ($M = 3.63$, $SD = 1.22$) than the authors of accepted reviews ($M = 3.40$, $SD = 1.33$). Fig. 2 shows the distribution of star ratings for both sub-samples.

The results of the preliminary study show the potential of

Table 1
Results for the preliminary study.

Information	Review status	M	SD	t	p
Identity-descriptive information					
Age	rejected	.35	.48	3.44	.001
	accepted	.38	.49		
Name	rejected	.37	.48	2.10	.035
	accepted	.39	.49		
Phone number	rejected	.18	.38	4.92	< .001
	accepted	.22	.41		
Identity disclosure level	rejected	.89	1.22	3.75	< .001
	accepted	.98	1.24		
Email provider type	rejected	.34	.47	11.50	< .001
	accepted	.45	.50		
Additional information					
Text length	rejected	149.20	241.61	19.96	< .001
	accepted	256.59	286.79		
Star rating	rejected	3.63	1.22	8.80	< .001
	accepted	3.40	1.33		

Abbreviations:

M = mean values, SD = standard deviation, t = t value, p = significance level.
Coding procedures:

Age: 1 = birth date provided, 0 = birth date not provided.

Name: 1 = name provided, 0 = no name or pseudo provided.

Phone number: 1 = phone number provided, 0 = no phone number provided.

Identity disclosure level: summated indicator of Age, Name, and Phone number.

Email provider type: 1 = telecommunication provider domain, 0 = else.

Text length = number of characters of review text.

Star rating = assigned star rating by the internet user between 0 and 5.

contextual indicators in assisting internet users in their navigation within online review sites. While the rejection/acceptance decision of the partner review site was solely affected by the review text itself, the significant differences in identity-descriptive characteristics underline the value of this indicator for review site operators. Furthermore, the results further motivate our research in investigating the value of contextual factors such as identity disclosure and star ratings from the internet user's perspective.

4. Main study

4.1. Experimental design and procedure

To test the formulated hypotheses, an experimental main study was conducted using a between-subjects factorial design. Previous empirical work highlights the relevance of online reviews for decision-making processes for services, usually high in experience qualities that are difficult to evaluate prior to one's own service experience (Sen and Lerman, 2007; Senecal and Nantel 2004). As a consequence, online reviews about a fictitious restaurant (i.e., Hugo's) published on a fictive review site (i.e., restocritics.com) were used as stimuli in both experiments. Recent findings that confirmed the role of review characteristics such as review length (Pan and Zhang, 2011, see results of the preliminary study in the present research), word use (Ott et al., 2011), and word duplication (Bhattarai, Rus and Dasgupta, 2009) on consumers' perceptions and the expected deceitfulness of the review were taken into account during the stimuli development to avoid unintended effects. Furthermore, close attention was paid to the realistic appearance and presentation of the stimuli (see Appendix 1 for two sample stimuli used in the experiment).

In the experiment, a $3 \times 2 \times 2$ between-subjects factorial design was adopted manipulating the three factors as follows: the

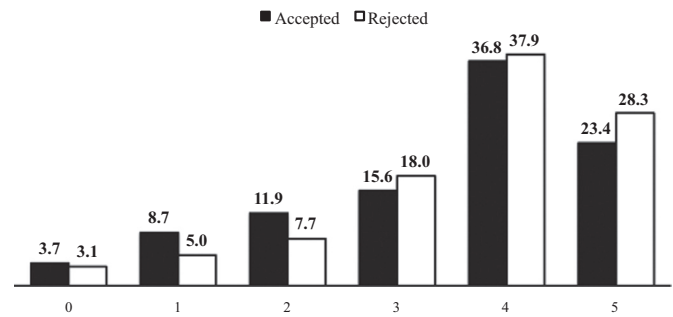


Fig. 2. Star ratings according to review status (in percent).

amount of information about the reviewer (high, moderate, low), consensus information (high consensus, low consensus), and the activation of persuasion knowledge (activation, no activation).

Two developed online survey questionnaires guided the participants through the studies. After a welcoming page, respondents were asked to answer some general questions regarding their familiarity with online reviews before they were exposed to the decision-making situation: *You are planning to go to a restaurant with your partner tonight and are faced with the decision to choose a cozy restaurant with good food.* The instruction then describes that the participant goes online to find the right place and further information on the restaurant on the review site restocritics.com.

Newspaper articles and reports about marketplace tactics present one source to activate persuasion knowledge of customers (Friestad and Wright, 1994). Consequently, the persuasion knowledge activation condition included a short news article *Lies and deception in consumer online reviews—read reviews critically* which describes that review sites are having a hard time detecting and eliminating fake reviews. In the next step, respondents were exposed to a screenshot of the review site, including the online review for Hugo's restaurant. The amount of disclosed information about the author of the review was manipulated. In the full disclosure condition, information on full surname and name, age, marital status, city and country and a photo of the review's author was provided. The first name, the initial of the surname, city and country as well as a photo of the reviewer's city were provided in the moderate disclosure condition. Finally, *user* and *unknown city* instead of names and place along with a generic replacement character for a photo were given in the low disclosure scenario. The consensus information was manipulated through the star ratings of Hugo's by the author of the review in question (5 out of 5 stars) compared to the current average of all reviewer ratings (4.5 stars for the high, 1.5 stars in the low consensus manipulations).

Participants then continued the survey with questions on their perceptions and the assessment of the dependent variables, trustworthiness, purchase intention, and avoidance behavior. Finally, participants were asked to evaluate how realistic the study context and stimuli were and to provide some socio-demographic information.

4.2. Measurement

Existing scales from the literature were adapted to the study context and respondents were asked to express their agreement with the statements on 7-point Likert response scales, where 1 represented *strongly disagree* and 7 *strongly agree*. The perceived trustworthiness of the source was adapted from Andrews et al. (1998) and included responses such as "In my opinion the source of the online review is trustworthy". Items from Lepkowska-White et al. (2003) were adapted to measure the respondent's purchase intention. The responses included statements such as "If I were

looking for a restaurant for tonight, my likelihood of going to the restaurant in the online review would be high". Responses such as "I will avoid making reservations at Hugo's restaurant in the future" were used to assess the respondent's intention to avoid the service provider and were adapted from [Graham Austin, Plouffe and Peters \(2005\)](#). Furthermore, the participant's familiarity with online reviews was included as potentially covariate. Responses such as "In general, I consider myself familiar with online reviews" were adapted from [Oliver and Bearden \(1985\)](#). To assess how realistic the scenarios and stimuli were, three items from [Bradley and Sparks \(2009\)](#) were included. Finally, two developed single-item statements were used to check for the successful manipulation of the amount of disclosed information on the reviewer and the consensus information. All used instruments exceeded conventional thresholds in terms of reliability and validity ([Hair et al., 2014](#)) (see Appendix 2 and 3).

4.3. Participants

The participants in the experimental study were recruited through Amazon Mechanical Turk (AMT). Participation in the study was proposed as a task on AMT and participants—so-called workers on AMT—were paid a small monetary compensation after filling out the questionnaire. AMT has received increasing attention as an alternative way to recruit participants for academic studies and recent findings provide support for the appropriateness of AMT for experimental studies ([Paolacci, Chandler and Ipeirotis, 2010](#); [Rand 2012](#)).

Respondents were randomly assigned to the conditions and the final samples consist of 390 participants equally distributed across experimental cells. The sample had a larger proportion of male participants (60.3%), and the average age of respondents was 30.44 years.

4.4. Results

Respondents perceived the scenario as believable ($M = 5.28$, $SD = 1.25$), found it easy to adopt the role of the customer ($M = 5.38$, $SD = 1.27$), and evaluated the given online review as realistic ($M = 5.20$, $SD = 1.39$). To assess the level of disclosed information on the reviewer's identity, respondents were asked to assess their perceived level of sufficient information about the author of the review. The manipulation worked out as intended, as the perceived amount of disclosed identity information was significantly higher in the high disclosure condition than in the moderate disclosure condition ($t = 29.432$, $p < .001$) and was significantly higher in the moderate disclosure condition than in the low disclosure case ($t = 29.011$, $p < .001$). The manipulation of the consensus of the review with the overall ratings worked as intended: the given review was perceived to be significantly more representative for the experiences by the multitude of internet users in the high consensus than in the low consensus condition ($t = 27.121$, $p < .001$). Respondents were familiar with online reviews ($M = 5.36$, $SD = 1.18$) and had moderate levels of trustworthiness ($M = 4.89$, $SD = 1.32$) and purchase intention ($M = 4.91$, $SD = 1.35$). The respondents' overall level of avoidance behavior towards the restaurant was rather low ($M = 2.90$, $SD = 1.65$).

An ANCOVA with the respondents' familiarity with online reviews as covariate was carried out to test the effects expected in the hypotheses. [Table 2](#) shows the results.

In H1, we expected that the amount of information disclosed by the author of the review impacts the perceived trustworthiness. We find a significant main effect of identity disclosure that supports H1 ($F = 32.097$, $p < .001$, $\eta^2 = .145$): the perceived trustworthiness of the source decreases with the reduction of available information about the reviewer ($M_{\text{high}} = 5.86$, $SD = 1.05$,

Table 2
ANCOVA results for the main study.

	F	η^2	M
Main effects			
Persuasion knowledge activation	1.107	.003	$M_{\text{prime}} = 4.70$ (.135) $M_{\text{no_prime}} = 5.06$ (1.27)
Consensus information	79.291***	.174	$M_{\text{high}} = 5.45$ (.81) $M_{\text{low}} = 3.90$ (1.36)
Identity disclosure	32.097***	.145	$M_{\text{high}} = 5.86$ (1.05) $M_{\text{moderate}} = 4.87$ (.96) $M_{\text{low}} = 3.87$ (1.33)
Two-way interaction effects			
Activation x Identity	.798	.004	
Activation x Consensus	.387	.001	
Consensus x Identity	2.463⁺	.013	High consensus: $M_{\text{high}} = 6.01$ (.78) $M_{\text{moderate}} = 5.15$ (.75) $M_{\text{low}} = 4.90$ (.96) Low consensus: $M_{\text{high}} = 5.00$ (1.79) $M_{\text{moderate}} = 4.24$ (1.07) $M_{\text{low}} = 3.39$ (1.20)
Three-way interaction effect			
Activation x Consensus x Identity	.289	.002	
Covariate			
Familiarity	.950	.003	

Mean values with standard deviations in brackets.

Abbreviations:

M = mean; η^2 = partial eta squared.

Significance levels:

* $p < .05$, ** $p < .01$.

⁺ $p < .1$.

*** $p < .001$.

$M_{\text{moderate}} = 4.87$, $SD = .96$, $M_{\text{low}} = 3.87$, $SD = 1.33$).

The results further support H2: consensus information significantly affects the perceived trustworthiness of the source ($F = 79.291$, $p < .001$, $\eta^2 = .174$). The respondent perceives the source as more trustworthy when consensus is high ($M = 5.45$, $SD = .81$) than when consensus is low ($M = 3.90$, $SD = 1.36$). Furthermore, the results support the interaction between the disclosure of identity information and the consensus information ($F = 2.463$, $p < .1$, $\eta^2 = .013$). Chart 3 in [Fig. 3](#) shows that while the negative effect of decreasing information on trustworthiness is somewhat smoothed in the high consensus condition, trustworthiness declines considerably with the reduction of available information on the reviewer's source in the low consensus case. This supports H3.

Finally, the activation of the persuasion knowledge does not significantly influence the perceived trustworthiness of the source ($F = 1.107$, $p > .1$), although the mean values seem to point in the expected direction. Consequently, H4 cannot be supported. The results in [Table 2](#) further show that the two-way interactions of the persuasion knowledge activation with identity disclosure ($F = .798$, $p > .1$) and consensus information ($F = .387$, $p > .1$), respectively are not significant.¹

To test the mediations formulated in hypotheses H5, H6, and H7 we used multiple ordinal least squares (OLS) regressions and 5000 bootstrap samples to construct 95% bias-corrected bootstrap confidence intervals (BCI) and to calculate the unstandardized path coefficients (i.e., B) and standard errors (SE) ([Hayes, 2013](#); [Preacher](#)

¹ The ANCOVA result on the non-significant two-way interactions of the persuasion knowledge activation were further tested using bootstrapping procedures in PROCESS ([Hayes, 2013](#)) with the same results: persuasion knowledge activation does not moderate the effects of identity disclosure and consensus information on trustworthiness.

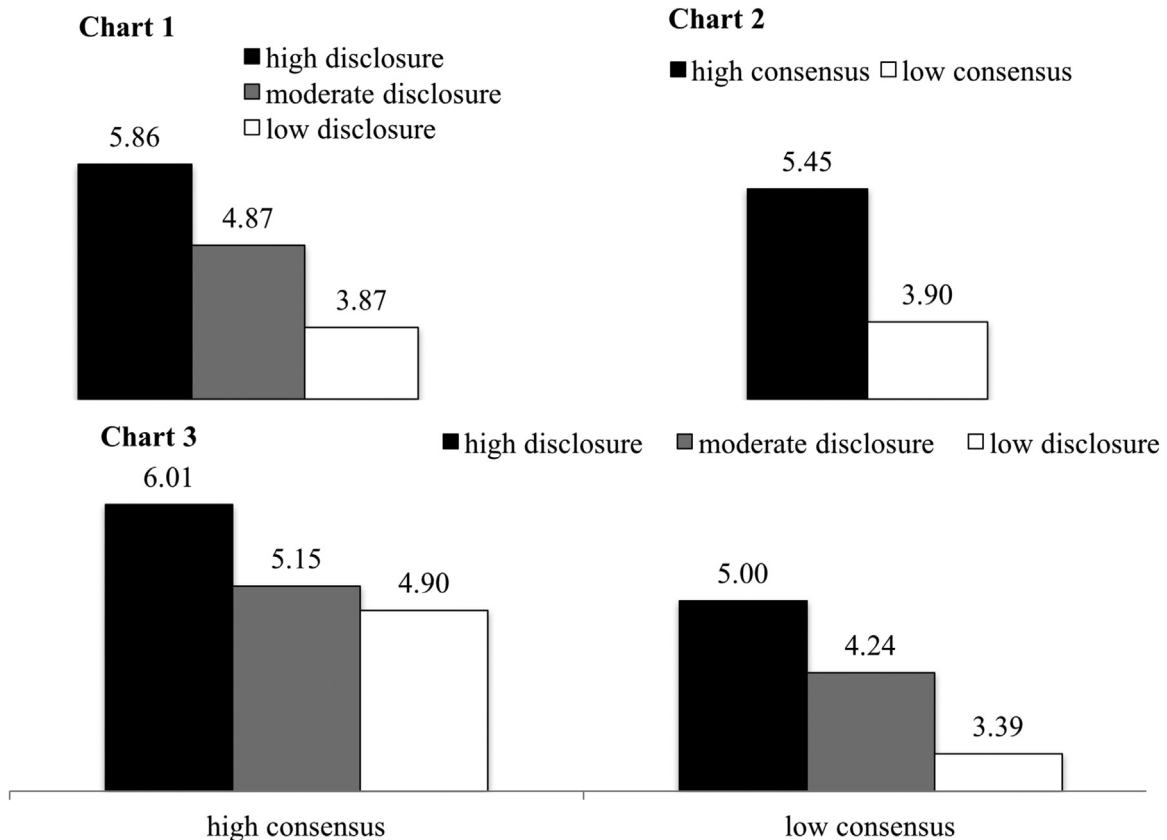


Fig. 3. The main effects of disclosed identity information (Chart 1), consensus information (Chart 2), and the interaction effect of identity disclosure and consensus (Chart 3) on perceived trustworthiness.

and Hayes 2008). Indirect effects tested through OLS regressions are considered as being significant when the 95% BCI excludes zero. The identity disclosure level indirectly affects purchase intention ($B = -.659$, $SE = .073$, $BCI_{low} = -.806$, $BCI_{up} = -.524$; H5a is supported) and avoidance behavior ($B = .269$, $SE = .076$, $BCI_{low} = .130$, $BCI_{up} = .426$; H5b is supported) through the perceived trustworthiness of the source: The perceived trustworthiness of the source decreases with the reduction of available information about the review's author ($B = -.979$, $p < .001$). Trustworthiness, in turn, significantly increases the intention to purchase from the service provider ($B = .673$, $p < .001$) and decreases the individual's intention to avoid the restaurant in the future ($B = -.275$, $p < .001$), respectively. The analysis of the direct effects reveal a slightly significant direct effect of the identity disclosure level on purchase intention ($B = -.130$, $p < .1$) and a significant and negative effect on avoidance behavior ($B = -.323$, $p < .05$).

The results show significant indirect effects of the consensus information on purchase intention ($B = -.954$, $SE = .109$, $BCI_{low} = -1.181$, $BCI_{up} = -.754$; H6a is supported) and intention to avoid the restaurant in the future ($B = .339$, $SE = .126$, $BCI_{low} = .105$, $BCI_{up} = .609$; H6b is supported) through trustworthiness: with the decrease of consensus, the review's source is perceived as less trustworthy ($B = -1.532$, $p < .001$), while trustworthiness positively affects purchase intention ($B = .623$, $p < .001$) and negatively influences intention to avoid the service provider ($B = -.242$, $p < .001$), respectively. The consensus information further has a significant direct effect on purchase intention ($B = -.455$, $p < .001$), but does not directly affect avoidance behavior ($B = -.242$, $p > .1$).

Finally, in hypothesis H7, we expected the effect of the persuasion knowledge activation on the dependent variables to be mediated by the perceived trustworthiness of the source. The results show significant indirect effects of the persuasion knowledge activation on purchase intention ($B = .262$, $SE = .095$, $BCI_{low} = .079$,

$BCI_{up} = .452$; H7a is supported) and intention to avoid the restaurant in the future ($B = -.046$, $SE = .032$, $BCI_{low} = -.130$, $BCI_{up} = -.001$; H7b is supported) through trustworthiness: when the respondent's persuasion knowledge is not activated (as compared to the deception-aware respondents), the review's source is perceived as more trustworthy ($B = .352$, $p < .05$), while trustworthiness positively affects purchase intention ($B = .711$, $p < .001$) and negatively influences intention to avoid the service provider ($B = -.126$, $p < .05$), respectively. Interestingly, the persuasion knowledge activation does not directly affect purchase intentions ($B = .075$, $p > .1$), but significantly and strongly influences avoidance behavior ($B = -.871$, $p < .001$).

5. Discussion

5.1. Summary and discussion of key findings

The present study aimed to assess empirically a number of contextual cues that potentially assist internet users in distinguishing fake from genuine reviews. While the literature often underlines the important complexity for consumers to detect deceptive opinion spam (Grazioli and Jarvenpaa, 2000; Mayzlin, Dover and Chevalier 2014), discussions of the topic in the literature and in the media require internet users to become skilled consumers of malicious practices such as fake reviews (George, Giordano and Tilley, 2016; George, Tilley and Giordano 2014). Recent surveys explain the difficulties of detection mechanisms in actually countering the professional production of authentic sounding fake reviews and highlight the important amount of fake reviews published on review sites (Streitfeld, 2012). In addition to a number of textual markers of deceptive opinion spam (Ott et al., 2011), the present study adopted a more customer-centric perspective and focused on

potential indicators that relate to the context of the review. Through a preliminary study of 9707 rejected and accepted online reviews, we were able to show that identity-related information significantly separates trustworthy from suspicious reviews for human-driven quality checks on a large review site. In an experimental main study, we then assessed the role of identity-descriptive information different as well as two additional types of information and mechanisms that potentially assist internet users in detecting fake reviews: in association with the increasing media coverage of deceptive reviews, one empirically tested factor dealt with the activation of a deception-aware mindset of the customer (Boush, Friestad and Wright, 2009). This aspect related to the persuasion knowledge of consumers (Friestad and Wright, 1994) was previously studied in similar contexts, but it yielded mixed results (Bambauer-Sachse and Mangold, 2013). In the present study, the priming condition that aimed to activate the deception-aware mindset of respondents shows some mixed results: the activation did not significantly affect the perceived trustworthiness of the source in the ANCOVA results, but had a significant effect when using unstandardized OLS path estimates. Furthermore, the persuasion knowledge activation did not significantly moderate the effects of identity disclosure and consensus information on the perceived trustworthiness. This finding, that requires looking with caution at the importance of persuasion knowledge activation, can potentially be explained by two aspects. First, the participants in the experiments considered themselves as rather competent and familiar users of online reviews. This factor might relate to a heightened awareness towards manipulative content within review sites, even without an immediate activation of a deception-aware mindset. Second, as the news article that was used in the priming condition was integrated prior to the two additional types of cues (i.e., identity disclosure and consensus), individuals relied much more on the immediate information that surrounded the given review and crowded out the potential effect of the persuasion knowledge activation itself. This rationale is consistent with Chaiken's (1980) heuristic approach and the expectation that individuals make less effort in assessing the validity of a piece of information. In the context of the present research, this relates to the role of the two additional and immediately accessible cues in screening out the impact of a preceding one (i.e., the priming condition).

The results across the two experiments, in turn, highlight the power of consensus—or social proof (Cialdini, 2007)—in peer-to-peer contribution-driven environments that such a review sites. In line with attribution theories (Kelley and Michela 1980; Weiner, 2000), the consistency of the review in question with the overall ratings by the review site's community helps users to attribute the causes for a positive review to the superior quality of the service than to a commercial bias of the reviewer (Mayzlin, Dover and Chevalier, 2014). Our findings are thus in line with previous studies that highlight the relevance of consensus information in online environments (Benedicktus, 2011; Benedicktus et al., 2010). In comparison with the focal cue—the amount of disclosed information on the reviewer's identity—high consensus compensates for missing information on the reviewer's identity.

The literature recently turned to the role of identity-disclosure in online contexts and supports the impact of the disclosure of identity information on reviews' helpfulness (Baek, Ahn and Choi, 2012; Ghose and Ipeirotis 2011) and credibility (Kusumasondjaja, Shanka and Marchegiani, 2012) as well as their impact on sales (Forman, Ghose and Wiesenfeld, 2008). Our findings corroborate those recent results and highlight the role of identity-disclosure—especially when additional information is scarce or ambiguous. Finally, the present study further underlines the relevance of trust-related factors in online environments. The source trustworthiness of a single review mediates the effects of available review cues on behavioral intentions directed at the service in question.

5.2. Theoretical contribution

This study and its findings contribute to the literature in various ways. First, across a preliminary study and an experimental main study, we highlight the importance of a number of available cues for internet users. Consequently, this study adopts a user- as opposed to a review-centric approach, addresses cues that affect the trustworthiness of a source and, thus, helps customers in detecting potential deceptive opinion spam. Furthermore, the research integrates three different types of information cues, namely the amount of disclosed information about the reviewer, the consensus between a review and the average ratings on a review site, and the activation of a deception-aware mindset. The preliminarily tested validity of one of the cues (i.e., identity-related disclosure) and the combination of this cue with two additional ones (i.e., consensus information and persuasion knowledge activation) in an experimental study reveal not only the general potential of two of those cues (Bambauer-Sachse and Mangold, 2013; Ghose and Ipeirotis 2011; Grazioli and Jarvenpaa 2000) but also the power of social proof (i.e., consensus information) and the amount of identity-related information about the reviewer in the trustworthiness assessment of a review's source by consumers. Second, the present study deepens the knowledge around the malicious practice of online marketplace deception in the marketing discipline (Boush, Friestad and Wright, 2009; Malbon, 2013) and, thus, complements related research in computer and information systems research (Bhattarai, Rus and Dasgupta, 2009; Ott et al., 2011). While the approaches in computer and information systems science are largely review-centric and should assist review site operators in updating and developing their filters for fake reviews, the present research is customer-centric and addresses information linked to the context in which the review is embedded (Luca and Zervas, 2014; Mayzlin, Dover and Chevalier 2014). The combination of review- and customer-centric approaches presents a valuable venue for the future processes to effectively address the matter. Third, our research contributes to the discussion of trust-related variables in anonymous online environments (Lee and Turban, 2001; Peppers and Rogers 2012). The present studies not only corroborate the crucial role of trustworthiness in stimulating internet users' purchasing behaviors in the short term, they also underline the potential consequences of a lack of trustworthiness in the medium and long term (i.e., avoidance behaviors). The perceived trustworthiness of the source of a given review was shown to affect those orthogonally directed behavioral intentions toward the reviewed service provider.

5.3. Managerial implications

The findings of our two studies have implications for the multiple stakeholders involved in the review ecosystem, namely service providers reviewed on those sites, review site operators themselves, and independent institutions and regulators. First, the significant effects of trustworthiness on immediate purchase intention and the future-oriented intention to avoid the service provider highlight the stakes involved in online reviews. The findings should therefore guide service providers to abandon and systematically refuse deceptive approaches such as fake online reviews. Online reputation firms usually directly approach service providers and offer their services. Although those firms may be sophisticated producers of authentic sounding review texts and might be able to circumvent review sites' filtering mechanisms, they are so far less skilled in manipulating the context in which the review will be embedded on the sites. As approaches that analyze the role of contextual factors in assisting internet users in detecting potential fake reviews should contribute in making users more skilled consumers of marketplace deception, the reputation firms' scope of action for malicious practices seems highly reduced. Consequently, as it is above all of the service provider in question (i.e., the restaurant in our research)

that seems affected by a lack of trustworthiness of a review's source, the provider should pass on the seductive power of fake reviews and embrace strategies that actively address responses to negative reviews (Dens, Pelsmacker and Purnawirawan, 2015; Sparks, So and Bradley 2016).

Second, the findings have important implications for the operators of review sites. The role of the identity-descriptive information asked and potentially required by review sites prior to the publication of a potential contributor seems a relevant direction for operators to reduce the risk of fake reviews. Review site operators frequently process submitted online reviews through a highly time-consuming and cost-intensive human quality check. Our results, in turn, propose two indicators—identity-related information and consensus information—that can potentially be integrated in automatically run quality checks of submitted reviews through algorithms. Furthermore, the presentation of a review and its consistency with the overall opinion of the community (i.e., the average star rating) should be prominent, such that readers of reviews can immediately infer the trustworthiness of the source from that information.

5.4. Limitations and further research

While the present research contributes to the literature and has valuable implications for various stakeholders in the ecosystem, we should address a number of potential limitations that lead to directions for future research. First, the present study concentrates on positive reviews and perceived trustworthiness. While intuitively the integration of potential deceptive positive reviews seems relevant—service providers might be interested in posting fake positive reviews to convince their potential customers—this

choice neglects the second facet of fake reviews: the publication of fake negative reviews in an aim to damage competitors. Recent cases, such as the condemnation of Samsung in Taiwan, which not only asked bloggers to publish positive evaluations of Samsung products but also paid them to post negative ones about their direct competitor HTC (Conner, 2013), highlight the importance of including negative fake reviews in future research. Some studies have also started to integrate both positive and negative reviews in their approaches (Bambauer-Sachse and Mangold, 2013; Mukherjee, Liu and Glance 2012).

Furthermore, the focus of the reviewer's identity information neglects the role of the reader's identity. Homophily or similarity between the sender and receiver/seeker of information has a long tradition in the WOM and interpersonal influence literature (Gilly et al., 1998; Wangenheim and Bayón 2004). Future studies should include the reader's profile and the match with the disclosed characteristics of the reviewer to investigate potential moderations of homophily in trustworthiness and subsequent behaviors (Naylor, Lamberton and Norton, 2011; Racherla, Mandviwalla and Connolly 2012).

Finally, two further aspects of the conducted research should be addressed: the chosen service context (i.e., restaurant) and the selected sampling method (i.e., AMT). As empirical studies suggest that WOM effectiveness varies across product categories (East, Hammond and Wright, 2007), future research should attempt to replicate our findings in additional categories. Furthermore, while online labor markets such as AMT present a valuable and robust way to access participants in academic research (Berinsky, Huber and Lenz, 2012; Goodman, Cryder and Cheema 2013; Rand 2012), future studies should adopt other recruitment strategies of participants (e.g., online panels) to expand our findings.

Appendix A. Sample stimuli

Sample stimulus used in the experiment: high identity disclosure, high consensus.

restocritics.com Search. Discover. Share With Others.

Search for: (Italian, French, ...) Where? (town, suburb, ...)

Restaurant reviews for: **HUGO's Restaurant** Overall rating: 131 reviews **4.5 / 5** Fotos (more...)

“Excellent for a charming night or a friendly moment!”
★★★★★ Reviewed August 23, 2012

Peter Muller
36 years, married
Strasbourg,
France

We go to Hugo's on a regular basis and went there again yesterday. The prices are really reasonable, considering the quality of the food. The menu is updated every month with the season and you can choose to match your wine with each course. It is very creative and tasty.

The nice filtered light offers a cozy atmosphere to have a charming dinner with your partner or to kick into the night a little bit later when diner time is over and when chef Hugo starts to serve cocktails.

Definitively a must see when you live in the city or you come to visit.

restocritics.com Search. Discover. Share With Others. About us Write a review Careers Site map Help center Public relations

Sample stimulus in the experiment: low identity disclosure, low consensus.



Appendix B. Quality of measurement instruments

	α	AVE	IR
Trustworthiness	.93	.87	
trustworthy			.94
credible			.93
believable			.93
Purchase intention	.92	.86	
If I were looking for a restaurant for tonight, my likelihood of going to the restaurant in the online review would be high.			.92
If I were to go to the restaurant, the probability that I would consider going to Hugo's restaurant would be high.			.93
If I had to make a reservation in a restaurant for tonight, my willingness to make the reservation at Hugo's restaurant would be high.			.93
Avoidance behavior	.90	.83	
I will avoid making reservations at Hugo's restaurant in the future.			.90
I disapprove of Hugo's restaurant due to the online review.			.95
I will boycott restaurants for which I read information like this one.			.88
Familiarity	.88	.80	
In general, I consider myself familiar with online reviews.			.89
... informed about online reviews.			.91
... knowledgeable about online reviews.			.89

Abbreviations: α = Cronbach's alpha; AVE = average variance extracted; IR = indicator reliability.

Appendix C. Descriptive statistics and discriminant validity of used constructs

	M	SD	(1)	(2)	(3)	(4)
(1) Trustworthiness	4.89	1.32	.94			
(2) Purchase intention	4.91	1.35	.71	.93		
(3) Avoidance behavior	2.90	1.65	–.17	–.31	.91	
(4) Familiarity	5.36	1.18	.13	.18	–.11	.90

M=mean values; SD=standard deviation; 1–4: correlations of constructs with the square roots of each construct's average variance extracted on the diagonal; AVE: average variance extracted. The square roots of the AVEs for the reflective constructs used in the studies are all higher than the correlations of these constructs with the other variables. In addition, the cross loadings were analyzed to check that all indicators load on the intended factors and that those loadings were higher than for all cross loadings. As all conditions were fulfilled, the discriminant validity was satisfied (Hair et al., 2014).

References

- Andrews, J., Craig, Netemeyer, Richard, G., Burton, Scot, 1998. Consumer Generalization of Nutrient Content Claims in Advertising. *J. Mark.* 62 (4), 62–75.
- Arndt, J., 1967. Role of Product-Related Conversations in the Diffusion of a New Product. *J. Mark. Res.* 4 (2), 291–295.
- Ba, Sulim, Pavlou, Paul A., 2002. Evidence of the Effect of Trust Building Technology in Electronic Markets: Price Premiums and Buyer Behavior. *MIS Q.* 26 (3), 243–268.
- Baek, Hyunmi, Ahn, JoongHo, Choi, Youngseok, 2012. Helpfulness of Online Consumer Reviews: Readers' Objectives and Review Cues. *Int. J. Electron. Commer.* 17 (2), 99–126.
- Bambauer-Sachse, Silke, Mangold, Sabrina, 2013. Do Consumers Still Believe What Is Said in Online Product Reviews? A Persuasion Knowledge Approach. *J. Retail. Consum. Serv.* 20 (4), 373–381.
- Bart, Yakov, Shankar, Venkatesh, Sultan, Fareena, Urban, Glen L., 2005. Are the Drivers and Role of Online Trust the Same for All Web Sites and Consumers? A Large-Scale Exploratory Empirical Study. *J. Mark.* 69 (4), 133–152.
- Benedicktus, R.L., 2011. The Effects of 3rd Party Consensus Information on Service Expectations and Online Trust. *J. Bus. Res.* 64 (8), 844–853.
- Benedicktus, R.L., Brady, M.K., Darke, Peter, Voorhees, C.M., 2010. Conveying Trustworthiness to Online Consumers: Reactions to Consensus, Physical Store Presence, Brand Familiarity, and Generalized Suspicion. *J. Retail.* 86 (4), 322–335.
- Berinsky, Adam J., Huber, Gregory A., Lenz, Gabriel S., 2012. Evaluating Online Labor Markets for Experimental Research: Amazon.Com's Mechanical Turk. *Political Anal.* 20 (3), 351–368.
- Bhattarai, A., V. Rus, D. Dasgupta (2009 Published), "Characterizing Comment Spam in the Blogosphere through Content Analysis," Conference Characterizing Comment Spam in the Blogosphere through Content Analysis.
- Bolton, Gary E., Katok, Elena, Ockenfels, Axel, 2004. How Effective Are Electronic Reputation Mechanisms? An Experimental Investigation. *Manag. Sci.* 50 (11), 1587–1602.
- Boush, D.M., Friestad, Marian, Wright, Peter, 2009. *Deception in the Marketplace*. Routledge, New York, NY.
- Bradley, Graham L., Sparks, Beverley A., 2009. Dealing with Service Failures: The Use of Explanations. *J. Travel Tour. Mark.* 26 (2), 129–143.
- Campbell, Margaret C., Kirmani, Amna, 2000. Consumers' Use of Persuasion Knowledge: The Effects of Accessibility and Cognitive Capacity on Perceptions of an Influence Agent. *J. Consum. Res.* 27 (1), 69–83.
- Campbell, Margaret C., Mohr, Gina S., Verlegh, Peeter W.J., 2013. Can Disclosures Lead Consumers to Resist Covert Persuasion? The Important Roles of Disclosure Timing and Type of Response. *J. Consum. Psychol.* 23 (4), 483–495.
- Chaiken, Shelly, 1980. Heuristic Versus Systematic Information Processing and the Use of Source Versus Message Cues in Persuasion. *J. Pers. Social. Psychol.* 39 (5), 752–766.
- Chatterjee, Patrali, 2001. Online Reviews: Do Consumers Use Them?. In: Gilly, M.C., Myers-Levy, J. (Eds.), *Advances in Consumer Research*. Association for Consumer Research, Valdosta, GA (Pp 129–34).
- Cialdini, R.B., 2007. *Influence: The Psychology of Persuasion*. Harpercollins, New York, NY.
- Colquitt, Jason A., Scott, Brent A., LePine, Jeffery A., 2007. "Trust, Trustworthiness, and Trust Propensity: A Meta-Analytic Test of Their Unique Relationships with Risk Taking and Job Performance. *J. Appl. Psychol.* 92 (4), 909–927.
- Cheryl Conner (2013). "Online Reputation: New Methods Emerge for Quashing Fake, Defamatory Reviews. (<http://www.forbes.com/sites/cherylsnappconner/2013/10/30/online-reputation-new-methods-emerge-for-quashing-fake-defamatory-reviews/>).
- Darke, Peter R., Ritchie, R.J.B., 2007. The Defensive Consumer: Advertising Deception, Defensive Processing, and Distrust. *J. Mark. Res.* 44 (1), 114–127.
- Dens, Nathalie, Pelsmacker, Patrick De, Purnawirawan, Nathalia, 2015. We(B)Care: How Review Set Balance Moderates the Appropriate Response Strategy to Negative Online Reviews. *J. Serv. Manag.* 26 (3), 486–515.
- East, Robert, Hammond, Kathy, Wright, Malcolm, 2007. The Relative Incidence of Positive and Negative Word of Mouth: A Multi-Category Study. *Int. J. Res. Mark.* 24 (2), 175–184.
- Eisend, Martin, 2004. Is It Still Worth to Be Credible? A Meta-Analysis of Temporal Patterns of Source Credibility Effects in Marketing. *Adv. Consum. Res.* 31 (1), 352–357.
- Forman, Chris, Ghose, Anindya, Wiesenfeld, Batia, 2008. Examining the Relationship between Reviews and Sales: The Role of Reviewer Identity Disclosure in Electronic Markets. *Inf. Syst. Res.* 19 (3), 291–313.
- Friestad, Marian, Wright, Peter, 1999. Everyday Persuasion Knowledge. *Psychol. Mark.* 16 (2), 185–194.
- Friestad, Marian, Wright, Peter, 1994. The Persuasion Knowledge Model: How People Cope with Persuasion Attempts. *J. Consum. Res.* 21 (1), 1–31.
- Friestad, Marian, Wright, Peter, 1995. Persuasion Knowledge: Lay People's and Researchers' Beliefs About the Psychology of Advertising. *J. Consum. Res.* 22 (1), 62–74.
- Gani, Aisha (2015). "Amazon Sues 1,000. 'Fake Reviewers'." (<http://www.theguardian.com/technology/2015/oct/18/amazon-sues-1000-fake-reviewers>). Accessed February 29, 2016.
- George, Joey F., Giordano, Gabriel, Tilley, Patti A., 2016. Website Credibility and Deceiver Credibility: Expanding Prominence-Interpretation Theory. *Comput. Human. Behav.* 54, 83–93.
- George, Joey F., Tilley, Patti, Giordano, Gabriel, 2014. Sender Credibility and Deception Detection. *Comput. Human. Behav.* 35 (0), 1–11.
- Ghose, Anindya, Ipeirotis, Panagiotis G., 2011. Estimating the Helpfulness and Economic Impact of Product Reviews: Mining Text and Reviewer Characteristics. *IEEE Trans. Knowl. Data Eng.* 23 (10), 1498–1512.
- Gilly, M.C., Graham, J.L., Wolfinbarger, M.F., Yale, L.J., 1998. A Dyadic Study of Interpersonal Information Search. *J. Acad. Mark. Sci.* 26 (2), 83–100.
- Goodman, Joseph K., Cryder, Cynthia E., Cheema, Amar, 2013. Data Collection in a Flat World: The Strengths and Weaknesses of Mechanical Turk Samples. *J. Behav. Decis. Mak.* 26 (3), 213–224.
- Graham Austin, Caroline, Plouffe, Christopher R., Peters, Cara, 2005. Anti-Commercial Consumer Rebellion: Conceptualisation and Measurement. *J. Target. Meas. Anal. Mark.* 14 (1), 62–78.
- Grazioli, S., Jarvenpaa, S.L., 2000. Perils of Internet Fraud: An Empirical Investigation of Deception and Trust with Experienced Internet Consumers. *IEEE Trans. Syst., Man., Cybern. - Part A: Syst. Hum.* 30 (4), 395–410.
- Grazioli, S., Jarvenpaa, Sirkka L., 2003. Deceived: Under Target Online. *Commun. ACM* 46 (12), 196–203.
- Gupta, Pranjal, Harris, Judy, 2010. How E-Wom Recommendations Influence Product Consideration and Quality of Choice: A Motivation to Process Information Perspective. *J. Bus. Res.* 63 (9/10), 1041–1049.
- Hair Jr., Joseph F., Thomas, G., Hult, M., Ringle, C.M., Sarstedt, Marko, 2014. *A Primer on Partial Least Squares Structural Equation Modeling (PLS-Sem)*. SAGE Publications, Thousand Oaks, CA.
- Hardesty, David M., Bearden, William O., Carlson, Jay P., 2007. Persuasion Knowledge and Consumer Reactions to Pricing Tactics. *J. Retail.* 83 (2), 199–210.
- Hayes, Andrew F., 2013. *Introduction to Mediation, Moderation, and Conditional Process Analysis: A Regression-Based Approach*. Guilford Press, New York, NY.
- Hovland, Carl I., Janis, Irving L., Kelley, Harold H., 1953. *Communication and Persuasion: Psychological Studies of Opinion Change*. Yale University Press, New Haven.
- Howell, Lee, 2013. *Global Risks 2013*. World Economic Forum, Geneva.
- Jiménez, Fernando R., Mendoza, Norma A., 2013. Too Popular to Ignore: The Influence of Online Reviews on Purchase Intentions of Search and Experience Products. *J. Interact. Mark. (Mergent, Inc.)* 27 (3), 226–235.
- Jindal, Nitin and Bing Liu (2008 Published), "Opinion Spam and Analysis," Conference Opinion Spam and Analysis. ACM. 219–230.
- Johnston, Katie (2013). "Review Websites Try to Thwart False Customer Ratings. (<http://www.bostonglobe.com/business/2013/09/24/fake-online-reviews-proliferate-fraud-detection-efforts-grow/aWsy3l5xQAFjxczMySMMKL/story.html>). Accessed January 15, 2014.

- Kaikati, Andrew M., Kaikati, Jack G., 2004. Stealth Marketing: How to Reach Consumers Surreptitiously. *Calif. Manag. Rev.* 46 (4), 6–22.
- Kelley, Harold H., Michela, John L., 1980. Attribution Theory and Research. *Annu. Rev. Psychol.* 31, 457–501.
- Kimmel, Allan J., 2010. Connecting with Consumers - Marketing for New Market Realities. Oxford University Press., Oxford.
- Kirmani, Amna, Zhu, Rui, 2007. Vigilant against Manipulation: The Effect of Regulatory Focus on the Use of Persuasion Knowledge. *J. Mark. Res. (JMR)* 44 (4), 688–701.
- Kusumasondjaja, Sony, Shanka, Tekle, Marchegiani, Christopher, 2012. Credibility of Online Reviews and Initial Trust: The Roles of Reviewer's Identity and Review Valence. *J. Vacat. Mark.* 18 (3), 185–195.
- Lee, Matthew K.O., Turban, Efraim, 2001. A Trust Model for Consumer Internet Shopping. *Int. J. Electron. Commer.* 6 (1), 75–91.
- Lepkowska-White, Elzbieta, Thomas, G. Brashear, Marc, G. Weinberger, 2003. A Test of Ad Appeal Effectiveness in Poland and the United States: The Interplay of Appeal, Product, and Culture. *J. Advert.* 32 (3), 57–67.
- Li, Fangtao, Minlie Huang, Yi Yang, and Xiaoyan Zhu (2011). "Learning to Identify Review Spam." in Twenty-Second International Joint Conference on Artificial Intelligence, Tobey Walsh ed. Barcelona, Spain: AAAI Press/International Joint Conferences on Artificial Intelligence, 2488–2493.
- Lim, Ee-Peng, Viet-An Nguyen, Nitin Jindal, Bing Liu, and Hady W. Lauw (2010 Published). "Detecting Product Review Spammers Using Rating Behaviors." Conference Detecting Product Review Spammers Using Rating Behaviors. ACM.
- Lorenzon, Kristian, Russell, Cristel Antonia, 2012. From Apathy to Ambivalence: How Is Persuasion Knowledge Reflected in Consumers' Comments About in-Game Advertising? *J. Mark. Commun.* 18 (1), 55–67.
- Luca, Michael and Georgios Zervas (2014). "Fake It Till You Make It: Reputation, Competition, and Yelp Review Fraud." in Harvard Business School NOM Unit Working Paper No. 14–006, (SSRN): (<http://ssrn.com/abstract=2293164>).
- Magnini, Vincent P., 2011. The Implications of Company-Sponsored Messages Disguised as Word-of-Mouth. *J. Serv. Mark.* 25 (4), 243–251.
- Malbon, Justin, 2013. Taking Fake Online Consumer Reviews Seriously. *J. Consum. Policy* 36 (2), 139–157.
- Masip, J., Garrido, E., Herrero, C., 2004. Defining Deception. *Anales Psicologica* 20 (1), 147–171.
- Mayer, Roger C., Davis, James H., Schoorman, F. David, 1995. An Integrative Model of Organizational Trust. *Acad. Manag. Rev.* 20 (3), 709–734.
- Mayzlin, Dina, Dover, Yaniv, Chevalier, Judith, 2014. Promotional Reviews: An Empirical Investigation of Online Review Manipulation. *Am. Econ. Rev.* 104 (8), 2421–2455.
- Morgan, Robert M., Hunt, Shelby D., 1994. The Commitment-Trust Theory of Relationship Marketing. *J. Mark.* 58 (3), 20–38.
- Mukherjee, Arjun, Bing Liu, and Natalie Glance 2012 (Published). Spotting Fake Reviewer Groups in Consumer Reviews, Conference Spotting Fake Reviewer Groups in Consumer Reviews. Association for Computing Machinery. 191–200.
- Naylor, Rebecca W., Lamberton, Cait Poyner, Norton, David A., 2011. Seeing Ourselves in Others: Reviewer Ambiguity, Egocentric Anchoring, and Persuasion. *J. Mark. Res.* 48 (3), 617–631.
- Nielsen (2013). "Under the Influence: Consumer Trust in Advertising." (<http://www.nielsen.com/us/en/newswire/2013/under-the-influence-consumer-trust-in-advertising.html>). Accessed April 10, 2014.
- Oliver, Richard L., Bearden, William O., 1985. Crossover Effects in the Theory of Reasoned Action: A Moderating Influence Attempt. *J. Consum. Res.* 12 (3), 324–340.
- Orvis, Bruce R., Cunningham, John D., Kelley, Harold H., 1975. A Closer Examination of Causal Inference: The Roles of Consensus, Distinctiveness, and Consistency Information. *J. Pers. Social. Psychol.* 32 (4), 605–616.
- Ott, Myle, Claire Cardie, and Jeff Hancock (2012 (Published)). "Estimating the Prevalence of Deception in Online Review Communities." Conference Estimating the Prevalence of Deception in Online Review Communities. 2187864: ACM. 201–210.
- Ott, Myle, Yejin Choi, Claire Cardie, and Jeffrey T. Hancock (2011 (Published)). "Finding Deceptive Opinion Spam by Any Stretch of the Imagination." Conference Finding Deceptive Opinion Spam by Any Stretch of the Imagination. 309–319.
- Pan, Yue, Zhang, J.C., 2011. Born Unequal: A Study of the Helpfulness of User-Generated Product Reviews. *J. Retail.* 87 (4), 598–612.
- Paolacci, G., Chandler, J., Ipeirotis, P.G., 2010. Running Experiments on Amazon Mechanical Turk. *Judgm. Decis. Mak.* 5 (5), 411–419.
- Pavlou, Paul A., Dimoka, Angelika, 2006. The Nature and Role of Feedback Text Comments in Online Marketplaces: Implications for Trust Building, Price Premiums, and Seller Differentiation. *Inf. Syst. Res.* 17 (4), 392–414.
- Peppers, Don, Rogers, Martha, 2012. Extreme Trust - Honesty as a Competitive Advantage. Portfolio/Penguin, New York, NY.
- Preacher, Kristopher J., Hayes, Andrew F., 2008. Asymptotic and Resampling Strategies for Assessing and Comparing Indirect Effects in Multiple Mediator Models. *Behav. Res. Methods* 40 (3), 879–891.
- Racherla, Pradeep, Mandviwalla, Munir, Connolly, Daniel J., 2012. Factors Affecting Consumers' Trust in Online Product Reviews. *J. Consum. Behav.* 11 (2), 94–104.
- Rand, D.G., 2012. The Promise of Mechanical Turk: How Online Labor Markets Can Help Theorists Run Behavioral Experiments. *J. Theor. Biol.* 299 (21), 172–179.
- Roberts, Daniel (2013). "Yelp's Fake Review Problem." (<http://tech.fortune.cnn.com/2013/09/26/yelps-fake-review-problem/>). Accessed January 15, 2014.
- Rotter, Julian B., 1967. A New Scale for the Measurement of Interpersonal Trust. *J. Pers.* 35 (4), 651–665.
- Rumbo, J.D., 2002. Consumer Resistance in a World of Advertising Clutter: The Case of Adbusters. *Psychol. Mark.* 19 (2), 127–148.
- Rushe, Dominic (2013). "Fake Online Reviews Crackdown in New York Sees 19 Companies Fined." (<http://www.theguardian.com/world/2013/sep/23/new-york-fake-online-reviews-yoghurt>). Accessed January 13, 2014.
- Schindler, R.M., Bickart, B., 2005. Published Word of Mouth: Referable, Consumer-Generated Information on the Internet." Pp 35–61 in Online Consumer Psychology: Understanding and Influencing Consumer Behavior in the Virtual World. In: Haugtvedt, C.P., Machleit, K.A., Yalch, R.F. (Eds.), Lawrence Erlbaum Associates, Mahwah, NJ.
- Schlosser, Ann E., White, Tiffany Barnett, Lloyd, Susan M., 2006. Converting Web Site Visitors into Buyers: How Web Site Investment Increases Consumer Trusting Beliefs and Online Purchase Intentions. *J. Mark.* 70 (2), 133–148.
- Sen, Shahana, Lerman, Dawn, 2007. Why Are You Telling Me This? An Examination into Negative Consumer Reviews on the Web. *J. Interact. Mark.* 21 (4), 76–94.
- Senecal, Sylvain, Nantel, Jacques, 2004. The Influence of Online Product Recommendations on Consumers' Online Choices. *J. Retail.* 80 (2), 159–169.
- Sparks, Beverley A., So, Kevin Kam Fung, Bradley, Graham L., 2016. Responding to Negative Online Reviews: The Effects of Hotel Responses on Customer Inferences of Trust and Concern. *Tour. Manag.* 53, 74–85.
- Streitfeld, D. (2012). "The Best Book Reviews Money Can Buy." (http://www.nytimes.com/2012/08/26/business/book-reviews-for-hire-meet-a-demand-for-online-raves.html?pagewanted=1&_r=2&partner=rss&emc=rss). Accessed February 11, 2013.
- Sun, Tao, Youn, Seounmi, Wu, Guohua, Kuntaraporn, Mana, 2006. Online Word-of-Mouth (or Mouse): An Exploration of Its Antecedents and Consequences. *J. Comput. -Mediat. Commun.* 11 (4), 1104–1127.
- Tsang, Alex S.L., Prendergast, Gerard, 2009. Is a "Star" Worth a Thousand Words? *Eur. J. Mark.* 43 (11/12), 1269–1280.
- Tuttle, Brad (2012). "Why You Shouldn't Trust Positive Online Reviews - or Negative Ones, for That Matter." (<http://business.time.com/2012/08/28/why-you-shouldnt-trust-positive-online-reviews-or-negative-ones-for-that-matter/>). Accessed February 5, 2013.
- Wangenheim, Florian von, Bayón, Tomás, 2007. The Chain from Customer Satisfaction Via Word-of-Mouth Referrals to New Customer Acquisition. *J. Acad. Mark. Sci.* 35 (2), 233–249.
- Wangenheim, Florian von, Bayón, Tomás, 2004. The Effect of Word of Mouth on Services Switching. *Eur. J. Mark.* 38 (9/10), 1173–1185.
- Wei, Mei-Ling, Fischer, Eileen, Main, Kelley J., 2008. An Examination of the Effects of Activating Persuasion Knowledge on Consumer Response to Brands Engaging in Covert Marketing. *J. Public Policy Mark.* 27 (1), 34–44.
- Weiner, Bernard, 2000. Attributional Thoughts About Consumer Behavior. *J. Consum. Res.* 27 (3), 382–387.
- Xiao, B., Benbasat, I., 2011. Product-Related Deception in E-Commerce: A Theoretical Perspective. *MIS Q.* 35 (1), 169–195.
- Xie, Sihong, Guan Wang, Shuyang Lin, and Philip S. Yu (2012 Published). "Review Spam Detection Via Temporal Pattern Discovery." Conference Review Spam Detection Via Temporal Pattern Discovery.
- Zhu, F., Zhang, X.Q., 2010. Impact of Online Consumer Reviews on Sales: The Moderating Role of Product and Consumer Characteristics. *J. Mark.* 74 (2), 133–148.